

Android VPN SDK

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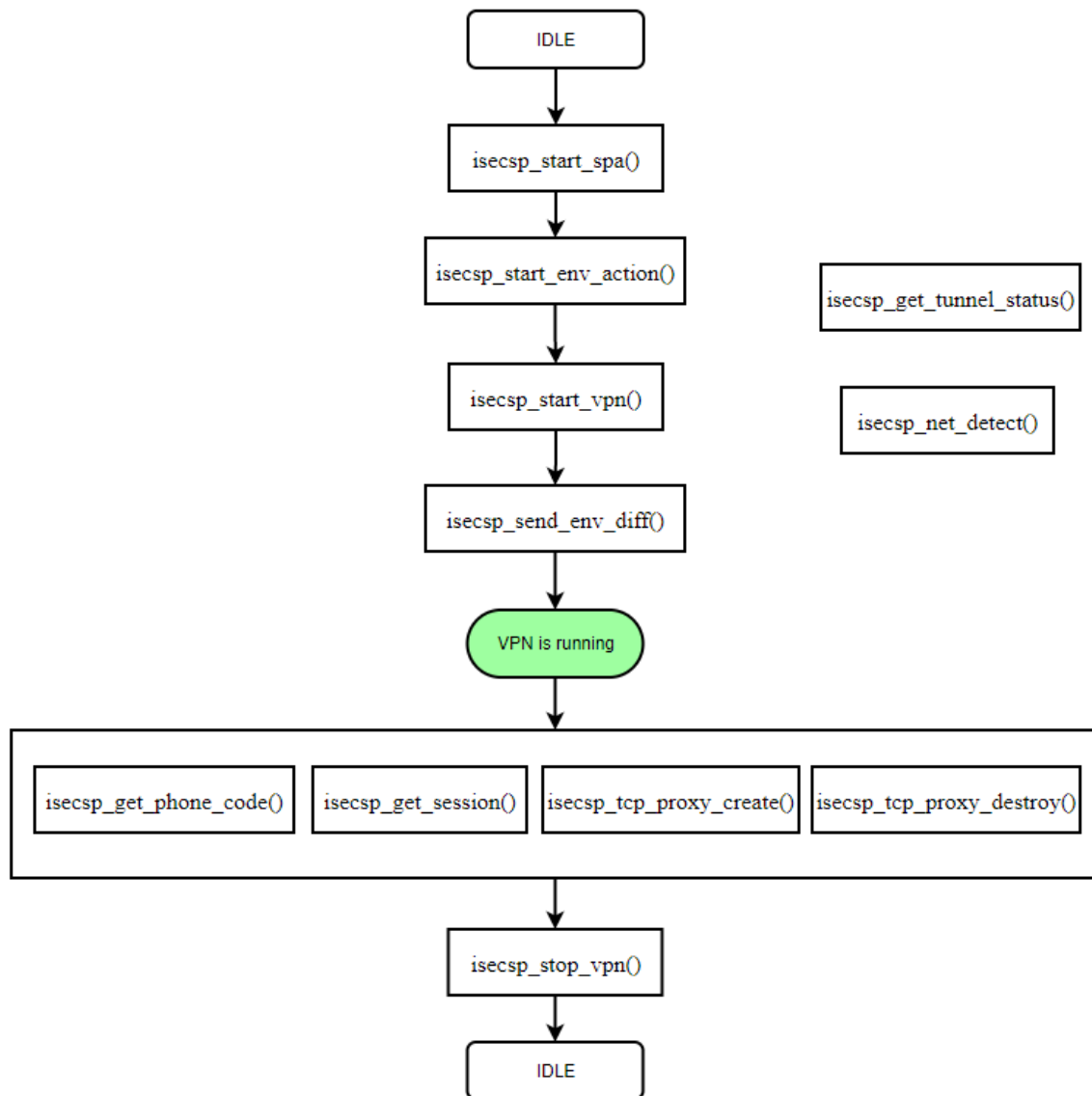
1. 综述

本文介绍面向 Android 平台的 VPN SDK 使用方法。其中包含 L3VPN 和 L4VPN 共有功能、特有功能，可根据需要选择使用。

VPN SDK 提供了一套简单易用的 API，帮助开发者在各平台上建立与 NETIAG 服务器相连接的 VPN 加密隧道。

VPN SDK 支持多种认证方式，包括单用户名密码认证，多因素认证，证书认证，设备 ID 认证，以及由上述各认证方式组合而成的多方法认证（不同平台支持的认证方式见下文详细描述）。SDK 提供接口支持用户输入各种认证凭证。只有认证通过，才能建立 VPN 隧道。

2. VPN 接口调用流程

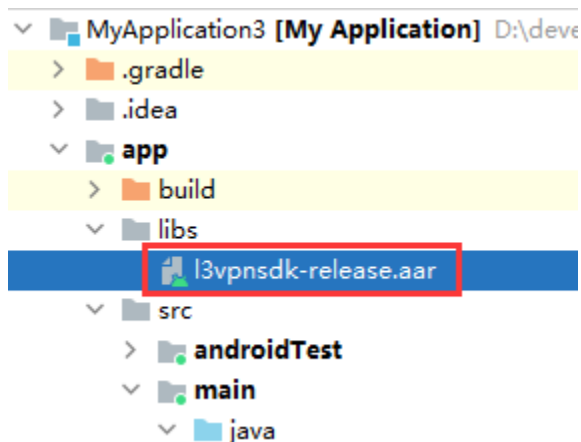


上图是 VPN 接口调用流程。

3. VPN SDK 集成

3.1 集成步骤

将 **l3vpnsdk-release.aar** 或 **l4vpnsdk-release.aar** 添加到工程的 **libs** 目录下，如下图



在项目的 **build.gradle** 文件中添加依赖

```
implementation files('libs/l3vpnsdk-release.aar')
```

3.2 VPN 相关接口

3.2.1 开启 SPA

```
int result = isecsp_start_spa(IsecspStartSPAPara para);
```

IsecspStartSPAPara 结构体说明：

参数名	类型	含义
username	String	SPA 用户名
userpass	String	SPA 密码
deviceid	String	设备唯一 ID

spa_port	int	SPA 端口
vpn_port	int	SPA Host
spahost	String	VPN 端口
vpnhost	String	VPN Host
enc_key	String	共享密码
timeout_value	int	连接超时时间
retry	int	重试次数
返回值	类型	含义
result	int	>0:成功 其他：失败

3.2.2 获取手机验证码

```
int result = isecsp_get_phone_code(IsecspGetPhoneCodePara para);
```

IsecspGetPhoneCodePara 结构体说明：

参数名	类型	含义
sms_url	String	获取手机验证码 url
phone_number	String	手机号
dynamic_code	String	动态码
返回值	类型	含义
result	int	0:成功 其他：失败

3.2.3 获取 Session

```
int result = isecsp_get_session(IsecspGetSessionPara para, CommonCallback callback);
```

IsecspGetSessionPara 结构体说明：

参数名	类型	含义
-----	----	----

host	String	VPN Host
port	int	VPN Port
username	String	VPN 登录用户名
password	String	VPN 登录密码 1
password2	String	VPN 登录密码 2
password3	String	VPN 登录密码 3
certPath	String	证书路径
certPass	String	证书密码
method	String	VPN 登录方法名
isMAuthLogin	boolean	是否使用 Mauth 证书登录
返回值	类型	含义
result	int	0:成功 其他：失败

3.2.4 启动 VPN

```
int result = isecsp_start_vpn(IsecspStartVPNPara para);
```

IsecspStartVPNPara 结构体说明：

参数名	类型	含义
host	String	VPN Host
port	int	VPN Port
username	String	VPN 登录用户名
password	String	VPN 登录密码 1
password2	String	VPN 登录密码 2
password3	String	VPN 登录密码 3
certPath	String	证书路径
certPass	String	证书密码

method	String	VPN 登录方法名
isMAuthLogin	boolean	是否使用 Mauth 证书登录
返回值	类型	含义
result	int	0:成功 其他：失败

注意：如果同时使用 L3VPN 和 L4VPN，需要先启动 L4VPN，再启动 L3VPN。

3.2.5 停止 VPN

```
int result = isecsp_stop_vpn(int logout);
```

参数名	类型	含义
logout	int	是否释放 session(1 是 0 否)
返回值	类型	含义
result	int	0:成功 其他：失败

3.2.6 获取 VPN 状态

```
int result = isecsp_get_tunnel_status();
```

返回值类型见 SDK 中 Common.VpnStatus 或本文档 [5.1 Common.VpnStatus](#)

3.2.7 设置 VPN 状态回调

设置 VPN 状态回调，如连接中/连接成功/连接失败等

```
VPNManager.initialize(this).set_callback(mHandler);
```

返回值 (what)	含义	备注
1	VPN 正在连接	无
2	VPN 连接成功	无
3	VPN 正在断开	无
4	VPN 已经断开	无
5	VPN 连接失败	可以通过 msg.getData().getInt(VpnMsg.MSG _VPN_ERROR_CODE)获取具体 的错误码，参考 demo 代码
7	VPN 正在重连	无
9	VPN 登陆回调	可以通过 IsecspLoginMethod methods = (IsecspLoginMethod) (msg.obj);获取具体的登陆方法， 参考 demo 代码
10	VPN 设备注册	无
11	VPN 环境检测	可以通过 String envConfig = (String) msg.obj;获取环境配置

3.2.8 开始环境检测

App 收到 VpnMsg.MSG_VPN_ENV 回调通知时，需要进行环境检测，调用函数：

```
int result = isecsp_start_env_action(CommonCallback callback, Context context);
```

环境检测结果在 callback 中回传

返回值	类型	含义
result	int	0:成功 其他：失败

3.2.9 发送环境检测 diff

将上次与本次环境检测结果进行对比，如果与上次不一致，生成 diff，调用此函数发送

```
int result = isecsp_send_client_rule_diff(String diff);
```

参数名	类型	含义
diff	String	环境检测后的差异
返回值	类型	含义
result	int	0:成功 其他: 失败

生成 diff 可参见：

```
CreateEnvDiffUtil.createDiff(lastEnvCheckResult, currentEnvCheckResult)
```

3.2.10 网络探测

```
String result = isecsp_net_detect(IsecspNetDetectPara isecspNetDetectPara);
```

IsecspNetDetectPara 结构体说明：

参数名	类型	含义
host	String	转换前 url 的 host
port	int	转换前 url 的端口
protocol	int	网络协议, 0(ICMP) 1(TCP)

返回值格式：

```
{"number": 2,  
  "ips": [
```

```
{  
  "rtt": 6,  
  "ip": "220.181.38.150"  
},  
{  
  "rtt": 7,  
  "ip": "220.181.38.149"  
}  
]  
}
```

在此之外的结果：

- (1) 返回值为空表示此 host:port 不通。
- (2) rtt<=0 表示此 host 或者对应的 IP 不通。

3.2.11 获取自定义认证字段返回值

```
String result = String getLoginReturn();
```

3.3 Mauth 相关接口

3.3.1 Mauth 接口调用流程

必要步骤:

```
isecsp_get_mauth_code() -> isecsp_initialization() -> isecsp_request_cert()
```

辅助功能:

```
isecsp_get_cert_field()  
isecsp_change_pin()  
isecsp_verify_pin()  
isecsp_is_locked()  
isecsp_unlock_user()  
isecsp_qr_login_by_cert()
```

3.3.2 获取注册码

```
String result = isecsp_get_mauth_code(String server, String username);
```

参数名	类型	含义
server	String	获取注册码 server 地址
username	String	用户名

返回值示例

获取注册码失败:

```
{"msg":"获取用户信息失败","resultcode":"E0000"}
```

获取注册码成功:

```
{"code":"cndstrw","resultcode":"000000"}
```

3.3.3 初始化 MAuth SDK

```
void isecsp_initialization(Context context, String server, String appld, String appSecret, String  
userName, String registerCode, String defaultPinCode, CommonCallback resultCallback)
```

参数名	类型	含义
context	Context	上下文环境
server	String	移动认证平台服务器地址
appID	String	应用 ID
appSecret	String	应用 Secret
userName	String	用户名
registerCode	String	注册码
defaultPinCode	String	默认 PIN 码
resultListener	ResultListener	初始化结果回调接口

3.3.4 下载证书

```
void isecsp_request_cert(@NonNull final String userName, @NonNull final String PIN, @NonNull  
final ResultListener resultListener);
```

参数名	类型	含义
username	String	用户名
PIN	String	证书 PIN 码
resultListener	ResultListener	结果回调接口

3.3.5 获取证书信息

```
String result = isecsp_get_cert_field(@NonNull String userName, int field);
```

参数名	类型	含义
username	String	用户名
field	int	要获取的证书字段 0.证书 DN 1.证书 SN 2.证书颁发者 3.保留字段 4.密钥用法 5.密钥长度 6.证书有效期开始 7.证书有效期结束 8.证书签名算法 9.证书版本
返回值	类型	含义
result	String	根据传入的字段，返回对应的值

3.3.6 修改 PIN 码

```
void isecsp_change_pin(@NonNull final String userName, @NonNull final String oldPIN,  
@NonNull final String newPIN, final ResultListener resultListener);
```

参数名	类型	含义
-----	----	----

username	String	用户名
oldPIN	String	旧 PIN 码
newPIN	String	新 PIN 码
resultListener	ResultListener	结果回调接口

3.3.7 校验 PIN 码

```
void isecsp_verify_pin(@NonNull String userName, @NonNull String PIN, @NonNull  
ResultListener resultListener);
```

参数名	类型	含义
username	String	用户名
PIN	String	要验证的 PIN 码
resultListener	ResultListener	结果回调接口

3.3.8 用户是否被锁

```
boolean result = isecsp_is_locked(String userName);
```

参数名	类型	含义
username	String	用户名
返回值	类型	含义
result	boolean	已经锁定返回 true，其他 false

3.3.9 解锁被锁用户

PIN 码有 10 次错误尝试次数，如果证书的 PIN 码连续错误输入次数达到 10 次，就会被锁定，锁定后的用户无法再使用需要 PIN 码验证的接口。

集成应用可以调用[判断用户是否被锁](#)接口获取当前用户的锁定状态。

用户一旦被锁定，如果需要使用，需要调用用户解锁接口进行解锁。

```
void isecsp_unlock_user(@NonNull final String userName, @NonNull String unLockCode,
    @NonNull final ResultListener resultListener);
```

参数名	类型	含义
username	String	用户名
unLockCode	String	解锁码
resultListener	ResultListener	结果回调接口

3.3.10 扫码登录

```
void isecsp_qr_login_by_cert(@NonNull final String userName, @NonNull String PIN, @NonNull
    String qrData, final int type, @NonNull final ResultListener resultListener);
```

参数名	类型	含义
username	String	用户名
PIN	String	证书 PIN 码
qrData	String	二维码数据
type	int	签名类型 0:raw 签名 1:attach 签名

		2:detach 签名
resultListener	ResultListener	结果回调接口

3.4 L3VPN 特有功能

3.4.1 添加白名单应用

支持添加白名单应用列表。如果列表中包含一个或多个应用，则只有列表中的应用才会使用 VPN。（不在列表中的）所有其它应用都将使用系统网络。如果列表为空，则所有应用都将使用 VPN。默认列表为空。

```
VPNManager.getInstance().add_allowed_application(new String[]{"packageName"});
```

3.4.2 SSL 优先协议设置

调用 `setSslProtocol()` 函数设置优先协议。

无证书时，默认为国密协议优先；

有证书时，根据证书类型自动选择。

```
public final int SSL_PROTOCOL_INIT = 0;
public final int SSL_PROTOCOL_TLS10 = 1;
public final int SSL_PROTOCOL_TLS12 = 2;
public final int SSL_PROTOCOL_SSLV3 = 3;
public final int SSL_PROTOCOL_SM2 = 4;
VPNManager.getInstance().set_ssl_protocol(VPNManager.getInstance().SSL_PROTOCOL_SM2)
;
```

3.5 L4VPN 特有功能

3.5.1 创建 TCP 端口

```
int result = isecsp_tcp_proxy_create(String host, int port);
```

L4VPN 已连接状态下，调用此接口打开一个端口(result)，使用 127.0.0.1:result 可打开对应地址。

参数名	类型	含义
host	String	原始 url 的 IP 或域名
port	int	原始 url 的端口
返回值	类型	含义
result	int	创建的端口

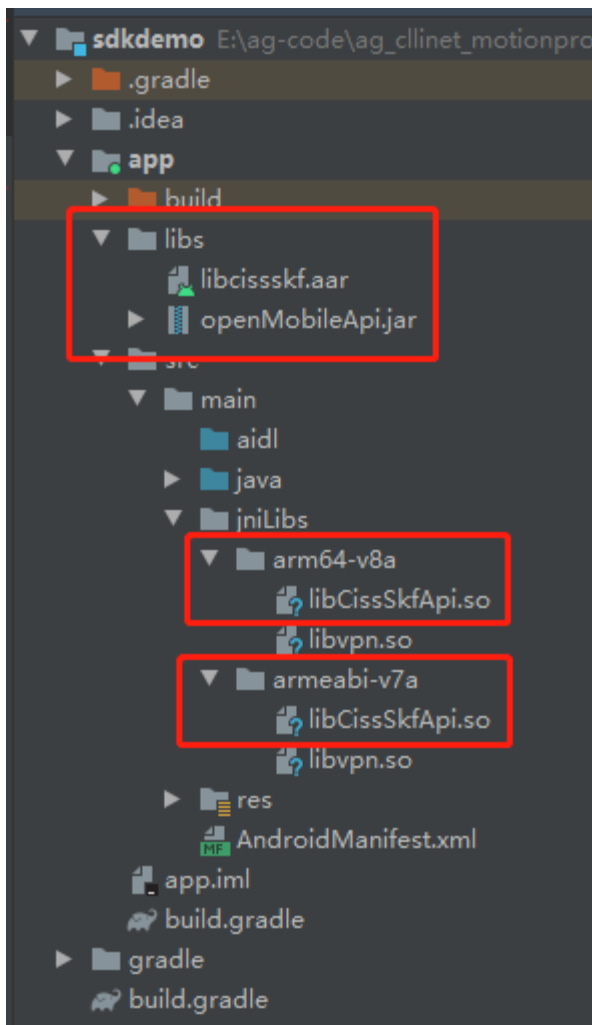
3.5.2 释放端口

```
int result = isecsp_tcp_proxy_destroy(int port);
```

参数名	类型	含义
port	int	要释放的端口
返回值	类型	含义
result	int	0:成功 其他：失败

3.6 SKF SDK 集成（无需 SKF 功能略过此步骤）

复制 SKF 库文件到项目 jniLibs 及 libs 文件夹，如下图：



关联所需依赖库

```
implementation 'org.bouncycastle:bcprov-jdk15on:1.62'  
implementation files('libs/libcisskf.aar')  
compileOnly files('libs/openMobileApi.jar')
```

初始化

```
// 代码中初始化
Application.onCreate()
SDKConfig.setStartSkf(true); //是否开启 skf 认证
boolean scanBleNow = false; //是否进行蓝牙扫描，否表示不进行蓝牙扫描
CISSSKFInit.init(this, scanBleNow); //初始化之后可以设置日志显示与否，默认不显示
OMAAPDUSenderForSKF.getInstance().setLogAble(true);
// 配置清单文件
<uses-permission android:name="android.permission.READ_EXTERNAL_STORAGE" />
<uses-permission android:name="android.permission.NFC" />
<uses-permission android:name="android.permission.SMARTCARD" />
<uses-permission android:name="org.simalliance.openmobileapi.SMARTCARD" />
<uses-permission android:name="android.permission.ACCESS_COARSE_LOCATION" />
<uses-library android:name="org.simalliance.openmobileapi"
android:required="false" />
```

具体方法解释

相关函数均声明在 VPNManager 中，调用前需导包：net.vpnsdk.vpn.VPNManager

```
// 枚举设备列表，返回设备列表字符串，以 “,” 分割
String simEnumDev();
// 枚举 Application 列表，返回 Application 列表字符串，以 “,” 分割
String simEnumApp(String devName);
// 枚举 Container 列表，返回 Container 列表字符串，以 “,” 分割
String simEnumContainer(String appName);
// 校验 pin 码，返回值 0 表示 pin 码正确
int simCheckPinKey(String pinKey)
// 设置 sim 信息
simSetInfo(String selectedDevName, String selectedAppName, String selectedContainerName,
String pinKey)
// Pin 码是否已经输入 返回值 0 表示 pin 码已经正确输入
isPinKey()
```

4. 其他

4.1 代码混淆

如果需要混淆代码，为了保证 sdk 的正常使用，需要在混淆文件中加上下面混淆规则：

```
-keep class net.vpnsdk.vpn.** { *; }
```

4.2 调试

使用 ADB 命令查看 SDK 的 Logcat 日志：

```
adb logcat
```

设置日志级别，默认为 INFO 级：

```
mVpnManager.setLogLevel(Common.LogLevel.LOG_INFO, 0);  
mVpnManager.setLogLevel(Common.LogLevel.LOG_DEBUG, 0);
```

4.3 兼容性

Android4.4 - Android12

4.4 常见错误码说明

Error Code	Error Message
------------	---------------

0	Success.
1	Failure.
2	The Seconn client fails to receive data through the SSL connection. Please check the network connection.
3	The Seconn client fails to send data through the SSL connection. Please check the network connection.
4	The Seconn client fails to establish the SSL connection. Please check the network connection and your certificate.
5	The server needs to verify the client certificate. Please specify a certificate file.
6	The session is invalid. Please log in again.
7	The session is expired. Please log in again.
8	The Seconn client fails to allocate the memory. Please terminate all processes and try again.
9	The Seconn client fails to establish the TCP connection. Please check the network connection.
10	The Seconn client fails to establish the TCP connection. Please check the network connection.
11	The Seconn client fails to receive data through the TCP connection. Please check the network connection.

12	The Seconn client fails to terminate the TCP connection. Please check the network connection.
13	The Seconn client fails to establish the TCP connection. Please check the network connection.
14	The Seconn client fails to establish the TCP connection. Please check the network connection.
15	The Seconn client fails to send data through the TCP connection. Please check the network connection.
16	The Seconn client fails to establish the TCP connection. Please check the network connection.
17	The item is not found.
18	The Seconn client fails to parse the cookie.
19	The Seconn client fails to perform the HTTP redirection.
20	The location header does not exist in the HTTP response header.
21	The Seconn client fails to establish the TCP connection. Please terminate all processes and try again.
22	The number of applications exceeds the limit. Please check the number of applications.
23	The size of the buffer is small.
24	Unknown host. Please check the hostname.

25	The program is still running.
26	The program is not running.
27	The Seconn client fails to connect to the new program. Please terminate all programs and try again.
28	The Seconn client fails to create a new program. Please terminate all processes and try again.
29	The Seconn client fails to establish the TCP connection. Please check the network connection.
30	The server configuration is invalid. Please check the server configuration.
31	The user has logged into another computer. Please log out from all computers and try again.
32	The certificate password is wrong. Please check the certificate password.
33	The Seconn client fails to parse the certificate file. Please check your certificate file and your permission.
34	The Seconn client fails to obtain the certificate file. Please check your certificate file and your permission.
35	The Seconn client fails to establish the TCP connection. Please terminate all processes and try again.
36	The Seconn client fails to establish the TCP connection. Please terminate all processes and try again.

37	Login fails. Please check your username and password.
38	The Seconn client fails to obtain the AAA method. Please check your virtual site configuration or network connection.
39	The Seconn client fails to configure the L3VPN tunnel. Please check your installation.
40	The Seconn client fails to obtain the configurations of L3VPN. Please check you virtual site configuration or network connection.
41	The operation has been terminated.
42	The parameter is invalid.
43	The Seconn client fails to establish the VPN connection.
44	The Seconn client fails to configure VPN.
45	The Seconn client fails to set up the speed tunnel.
46	The Seconn client fails to send the UDP packet. Please check the network connection.
47	The Seconn client fails to receive the UDP packet. Please check the network connection.
48	The tun device is down.
49	The VPN service is not supported. Please contact your IT administrator.
50	The server fails to respond. Please check the network connection.
51	Client Security is not supported. Please contact your IT administrator.

52	Too many methods! Please contact your IT administrator.
53	The method is not supported. Please contact your IT administrator.
54	The Seconn client fails to terminate the TCP connection.
55	The Seconn client fails to start the HTTP proxy.
56	The Seconn client fails to start the TCP proxy.
57	The Seconn client fails to stop the HTTP proxy.
58	The Seconn client fails to stop the TCP proxy.
59	The Seconn client fails to stop the HTTP proxy and TCP proxy.
60	Login fails. Please check your input.
61	Your device has not been approved yet. Please contact your IT administrator.
62	Your device has been denied. Please contact your IT administrator.
63	The number of users for this device has reached the upper limit. Please contact your IT administrator.
64	The number of devices for a user has reached the upper limit. Please contact your IT administrator.
65	Your device has not been registered yet. Please register it.
66	The Seconn client fails to create a session.
67	Certificate Error. Please select a valid certificate.
68	Certificate Error. The signature of the client certificate is invalid.

69	Certificate Error. The client certificate is untrusted.
70	Certificate Error. The client certificate has expired.
71	Certificate Error. The client certificate is invalid.
72	Certificate Error. The client certificate has been revoked.
73	Certificate Error. The server certificate is required.
74	Certificate Error. The signature of the server certificate is invalid.
75	Certificate Error. The server certificate is untrusted.
76	Certificate Error. The server certificate has expired.
77	Certificate Error. The server certificate is invalid.
78	Certificate Error. The server certificate has been revoked.
79	Change password
80	Authorization fails. Please contact your administrator to obtain a valid identity.
81	Connection timeout.
82	Proxy authentication fails. Please try again.
83	The Seconn client fails to get your phone number to do OTP authentication.
84	The Seconn client fails to send the message to your phone.
85	Login fails. You have exceeded the number of SMS verification code attempts allowed.
86	The program fails to communicate. Please try again.

87	Your device has been denied. Please contact your administrator.
88	Your device has not been approved yet. Please contact your administrator.
89	The Seconn client fails to download the client security component. Please try again.
90	The Seconn client fails to modify the file. Please try again.
91	The Seconn client fails to open the certificate file.
92	The Seconn client fails to read the certificate file.
93	The client verification fails.
94	The local system does not pass the Client Security check.
95	The Seconn client need update.
96	The post url does not exist in the HTTP response.
97	The realm value does not exist in the HTTP response.
98	The server fails to return a correct response after a GET request is sent to the Login page.
99	The server fails to return a correct response after a POST request is sent to the Start page.
100	The csrf value does not exist in the HTTP response.
101	The dynamic data does not exist in the HTTP response.
102	The server fails to return a correct response after a GET request is sent to the Grid page.

103	The server fails to return a correct response after a POST request is sent to the Grid page.
104	The server fails to return a correct response after a GET request is sent to the Base64 page.
105	The secgrid value does not exist in the HTTP response.
106	The HTTP response fails to be Decoded.
107	The server fails to return a correct response after a POST request is sent to the Secgrid page.
108	Incorrect HTTP response! Please check whether the Syferlock authentication is supported on your site.

4.5 常见问题解决办法

1. VPNManager.initialize(this); 参数 Context 是 activity 还是必须要 ApplicationContext ?

答案: ApplicationContext 。

2. SPA 登录是可选的吗 ?

答案: 是的, NETIAG 可以配置开启或关闭。

3. 调用 isecsp_start_spa()函数怎么获取结果 ?

答案: 可以根据返回值进行判断, 返回值大于 0 表示成功, 反之失败。

4. start_vpn()函数中参数证书路径和证书密码是必传的吗？

答案：不是必传的，NETIAG 开启证书认证后需要传这两个参数。

5. SPA 登录是可选的吗？

答案：是的，NETIAG 可以配置开启或关闭。

6. SDK 返回错误信息“Please check your username and password .”？

答案：检查用户名和密码、动态码或者短信验证码是否输入正确。

5. 附录

5.1 Common.VpnStatus

```
public final static class VpnStatus {  
    /**  
     * The VPN is idle.  
     */  
    public final static int IDLE = 0;  
    /**  
     * The client is connecting to a server.  
     */  
    public final static int CONNECTING = 1;  
    /**  
     * Log in successfully and the VPN tunnel is established.  
     */  
    public final static int CONNECTED = 2;  
    /**  
     * The client is disconnecting from a server.  
     */  
    public final static int DISCONNECTING = 3;  
    /**  
     * The client is disconnected and is reconnecting to the server automatically.  
     */  
    public final static int RECONNECTING = 4;  
}
```